

BRAINPOWER
ROMANIAN KINDER



Teacher Code :

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SOLUTIONS

PHYSICS UNIT 2

Test theme: PEPPA PIG

Reading Time: 5 minutes
Writing time: 45 minutes

QUESTION AND ANSWER BOOK

<i>Number of questions</i>	<i>Number of questions to be answered</i>	<i>Number of marks</i>
6	6	50

- Students are permitted to bring into the test room: pens, pencils, highlighters, erasers, sharpeners, rulers, 1 Scientific calculator, 1 A4 page of notes
- Students are NOT permitted to bring into the examination: blank sheets of paper, white out liquid/tape or a CAS calculator.

Materials supplied

- Question and answer book.
- Working space is provided throughout the book.

Instructions

- Write your **name** in the space provided above on this page.
- All written responses must be in English or Romanian.

Students are NOT permitted to bring mobile phones and/or any other unauthorised electronic devices into the test room.

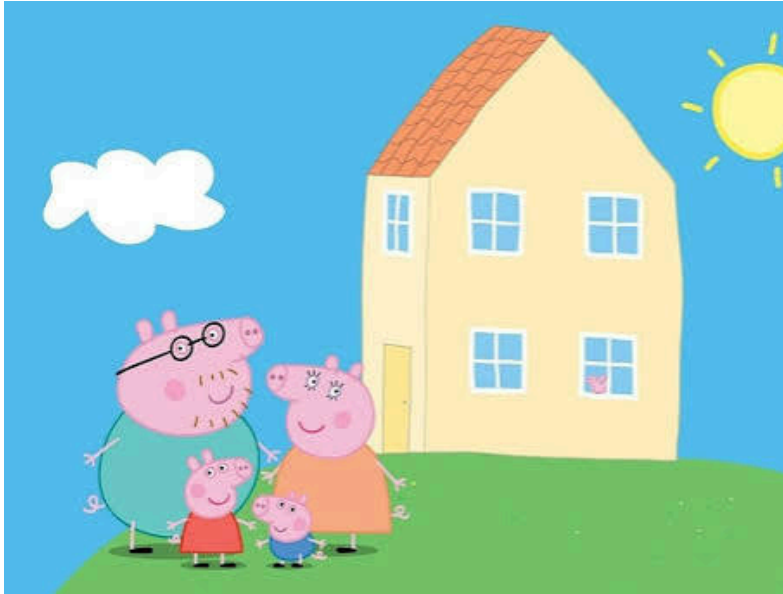
Instructions

Answer **all** questions in the spaces provided.

Unless otherwise specified an **exact** answer is required to a question.

In questions where more than one mark is available, appropriate working **must** be shown.

Gravity (g) = 9.81ms^{-2}

**Question 1** (10 marks)

The kitchen smells of ozone and rotting ham. Peppa is flickering in and out of existence. Her movement isn't linear; it follows a terrifying mathematical curve. You notice that as she moves across the floor, the tiles crack under her fluctuating weight. You need to predict her momentum before she 'stutters' into your personal space.

- a. Peppa, who is 120kg moves in a straight line towards you. Her motion is described by a velocity-time graph

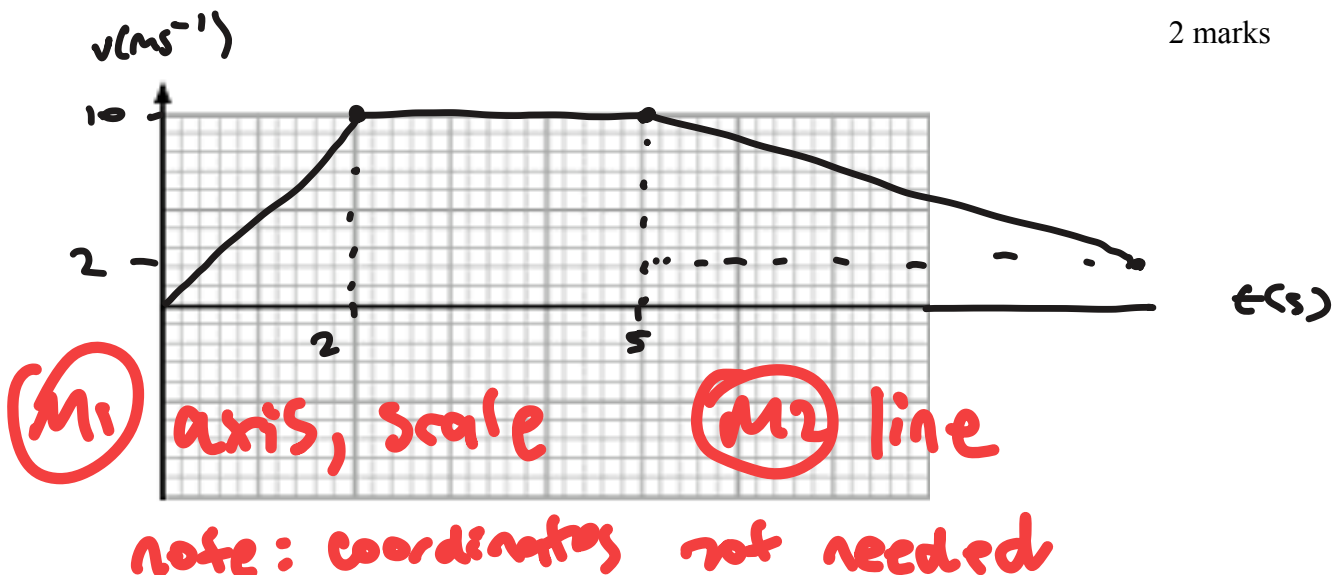
Phase A - Constant acceleration from rest to 10ms^{-1} over 2.0s

Phase B - Instantaneous glitch, she maintains 10ms^{-1} for 3.0s

Phase C - Constant deceleration to 2.0ms^{-1} over the next 5.0s

- i. Plot the velocity-time graph over an appropriate domain.

2 marks



- ii. Hence, calculate the total displacement Peppa covers during the entire 10.0 s interval.

1 Mark

$$\textcircled{1} 0-2s : \frac{2 \cdot 10}{2} = 10m$$

$$\textcircled{2} 2-5s : 10 \cdot 3 = 30m$$

$$\textcircled{3} 5-8s : \frac{8 \cdot 5}{2} + 10 = 30m$$

$$\therefore s = 70m$$

$$\textcircled{A1} s = 70m$$

- b. Calculate the magnitude of the net force acting on Peppa during Phase C.

2 Marks

$$F_{net} = ma$$

$$a = \frac{v-u}{t} = \frac{-8}{5} = -1.6ms^{-2}$$

$$m = 120kg$$

$$F_{net} = 120 \times -1.6 = -192N$$

$$\therefore \text{mag } F_{net} = 192N$$

$$\textcircled{M1} a = 1.6ms^{-2}$$

$$\textcircled{A1} F_{net} = 192N$$

- c. You throw a 3.0kg 'George' doll at her. It hits her at 20ms⁻¹ and rebounds at

10ms⁻¹. Calculate the average force exerted on the doll if the contact time is 0.04s

3 Marks

$$F_{at} = mv - mu$$

$$u = 20ms^{-1} \quad v = -10ms^{-1}$$

$$m = 3.0kg \quad t = 0.04s$$

$$F = \frac{3(-30)}{0.04} = -2250N$$

$$\textcircled{M1} \Delta v = 20 - (-10) = 30ms^{-1}$$

$$\textcircled{M1} F = \frac{m(v-u)}{\Delta t}$$

$$\textcircled{A1} F = 2250N$$

- d. Explain, using the concept of Impulse, why Peppa's head snaps back less if she 'glitches' (increases her mass) at the exact moment of impact

Impulse is equal to the change in momentum. 2 Marks

If Peppa's mass during contact increases, with impulse staying the same, her Δv must decrease, thus her head undergoes a smaller change.

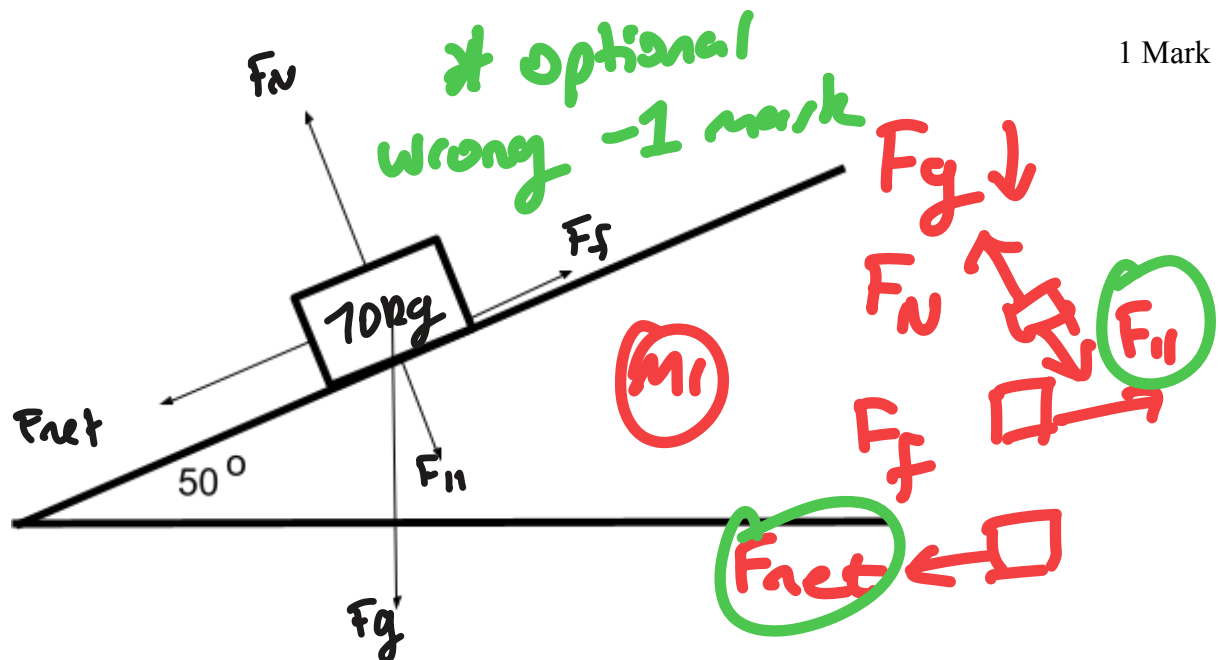
(M1) $I = m\Delta v$ (M2) inertia

Question 2 (8 Marks)

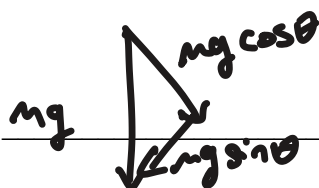
70kg Conseq if $m = 120\text{kg}$

You smash through the glass. The hill outside has shifted. It is now a jagged 50° slope of frozen, black 'muddy puddle' matter. The friction isn't constant, it's a physical resistance against your very soul. You need to calculate your descent before she 'falls' with terminal intent.

- a. Draw the force vectors on the diagram:



- b. Calculate the component of the gravitational force acting perpendicular to the slope



$m = 120\text{kg}$

$mg \cos \theta = 120 \times 9.81 \times \cos(50^\circ)$
 $= 441\text{ N}$

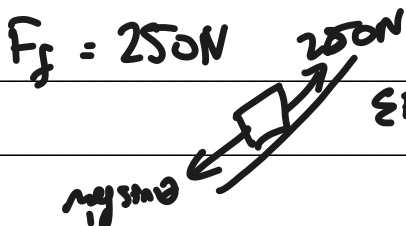
1 Mark

(M1) F_N or $F_{||}$
 $= 441\text{ N}$

- c. The normal force is found, and the mud provides a kinetic frictional force of 250N.

Calculate your net acceleration down the slope

2 Marks

$F_f = 250\text{N}$

 $\Sigma F = ma \Rightarrow F_D - F_f$
 $= 70 \times 9.81 \sin(50) - 250$
 $70a_1 = 276$
 $\therefore a_1 = 3.94\text{ms}^{-2}$
 down the slope.

(M1) F_{net}

(A1) $a = 3.94\text{ms}^{-2}$

- d. If the slope is 40m long, and you have an initial velocity of 2.0ms^{-1} (from the jump), calculate your velocity at the bottom.

2 Marks

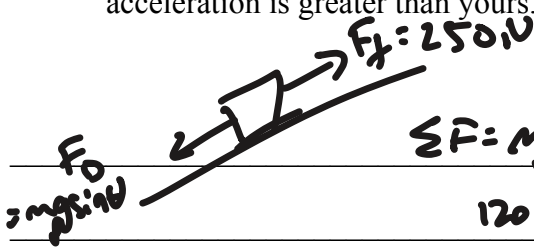
$a = 3.94\text{ms}^{-2}$, $u = 2.0\text{ms}^{-1}$, $v = ?$, $s = 40\text{m}$
 $v^2 = u^2 + 2as$
 $v = \sqrt{4 + 2(3.94)40} = \sqrt{319.2}$
 $\therefore v = 17.9\text{ms}^{-1}$ down the slope.

(M1) SUVAT

(A1) $v = 17.9\text{ms}^{-1}$ down slope

- e. Peppa (120kg) follows. If she experiences the same frictional force, show that her acceleration is greater than yours.

2 Marks


 $\Sigma F = ma_p \Rightarrow F_D - F_f$
 $120a_p = 120 \times 9.81 \sin(50) - 250$
 $\therefore a_p = 5.43\text{ms}^{-2}$
 $\therefore a_p > a_1$
 as $m_p > m_1$

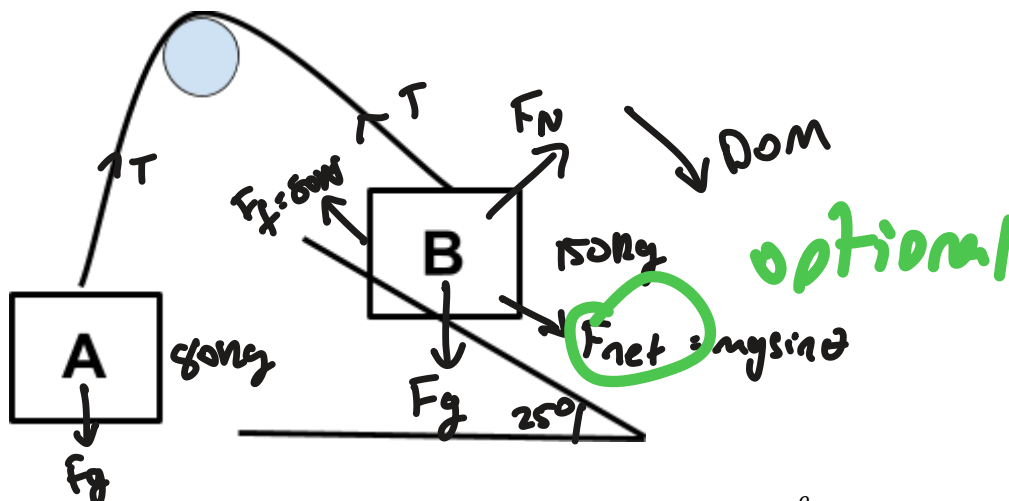
(M1) $a_p = 5.43\text{ms}^{-2}$

(M1) conclusion



Question 3 (9 Marks)

At the base, you find a meat-hook pulley system used for 'Grandpa Pig's Butcher Shop.' You strap yourself into a harness (Mass A). A heavy counterweight (Mass B) is sitting on a 25° ramp, connected to you via a cable over a pulley. This is your only way up to the attic escape



Mass A is 80kg hanging vertically, Mass B is 150kg on a 25° slope. The slope for B is not frictionless; it has a frictional force 80N . Mass B is released and slides down the slope.

- a. Draw the forces in the diagram above, including the direction of motion.

2 Marks

m_1 m_A

m_2 m_B

give mark if close both

b. Write the two equations of motion for Mass A and Mass B.

① $\Sigma F = m_A a = T - m_A g \quad | \quad 80a = T - 785$ (M₁) 2 Marks

② $\Sigma F = m_B a = m_B g \sin \theta - F_f - T \quad | \quad 150a = 622 - 80 - T$
 $150a = 542 - T$ (M₂)

* 1 mark if left in variables

c. Calculate the acceleration of the system

① + ②: $230a = T - 785 + 622 - 80 - T$ 2 Marks

$230a = -243$

$a = -1.06 \text{ ms}^{-2}$ down slope.

(M₁) $230a = -243$ (A₁) $a = -1.06$ down slope

d. Calculate the Tension (T) in the cable.

$80a = T - 785$

$785 - 85.2 = T$

$\therefore T = 700 \text{ N}$

-1 mark not included

2 Marks

(M₁) working

(A₁) $T = 700 \text{ N}$

e. If the cable has a maximum breaking strain of 900N, will it snap during this acceleration? Justify.

1 Mark

will not break as $F_s < T$,

where T is 700N less than F_s .

(M₁)

• NO

• $700 \text{ N} < 900 \text{ N}$



Question 4 (8 Marks)

The harness reaches the top. A steel bulkhead blocks the exit. A massive industrial spring (the 'Emergency Slaughter-Spring') is mounted to a piston. You must compress it to launch a 60kg into the door.

The spring constant $k = 15,000\text{Nm}^{-1}$

- a. You compress the spring by 0.75m . Calculate the energy stored

2 Marks

$$x = 0.75\text{ m}$$

$$U_e = \frac{1}{2} k x^2 = \frac{1}{2} (15,000) (0.75^2) \quad (M_1) \quad \text{Sub}$$

$$= 4219\text{ J} \quad (A_1)$$

- b. The piston is inefficient. As it launches the 60kg anvil, 15% of the stored energy is lost to heat/sound. Calculate the velocity of the anvil as it leaves the piston

3 Marks

$$\eta = \frac{\text{useful out}}{\text{total in}} = 15\% \quad (M_1)$$

$$E_k = \frac{1}{2} m v^2 \quad E_k = 85\% \text{ of } U_e = 3586\text{ J}$$

$$v = \sqrt{\frac{2E_k}{m}} = \sqrt{\frac{2 \cdot 3586}{60}} = 10.9\text{ ms}^{-1} \text{ in the DOM.} \quad (A_1)$$

- c. If the anvil travels 5.0m horizontally across a floor with a constant friction of 100N before hitting the door, calculate the Kinetic Energy of the anvil at the moment of impact.

#

3 Marks

(M)

$$E_{ki} = 3586 \text{ J}$$

$$W_f = F_s = 100 \times 5 = 500 \text{ N}$$

$$E_k = 3586 - 500$$

$$= 3086 \text{ N}$$

(A2)

Question 5 (6 Marks)

You (70kg) are moving East at 8.0 ms^{-1} . Peppa (120kg) is moving West at 4.0 ms^{-1} .

→ 8ms⁻¹← 4ms⁻¹

- a. After the collision, you bound back (West) at 1.0 ms^{-1} . Calculate Peppa's final velocity

3 Marks

$$\Sigma p_i = \Sigma p_f \rightarrow \text{Don't}$$

$$m_i v_i + m_p v_p = m_i v_{if} + m_p v_{pf}$$

$$70(8) + 120(-4) = -70 + 120 v_{pf}$$

$$v_{pf} = 1.25 \text{ ms}^{-1} \text{ East}$$

$$(M) p_i = 8012 \text{ ms}^{-1}$$

(M) working

$$(A) 1.25 \text{ ms}^{-1} \text{ E}$$

- b. Calculate the change in Kinetic Energy of the system

3 Marks

$$\Delta E_K = E_{Ki} - E_{Kf}$$

$$E_{Ki} = \frac{1}{2}(70)(64) + \frac{1}{2}(120)(16)$$

$$= 3200 \text{ J} \quad (M1)$$

$$E_{Kf} = \frac{1}{2}(70)(1) + \frac{1}{2}(120)(1.25^2)$$

$$= 129 \text{ J} \quad (M1)$$

$$\therefore \Delta E_K = 3071 \text{ J} \quad (A1)$$

- c. Is momentum conserved? Is energy conserved? Explain why these two answers might differ in a real-world scenario.

(M1) $p \checkmark$, $E \times$ (M2) closed body not possible 2 Marks

momentum is conserved, but energy is transferred into static electricity in the ground.

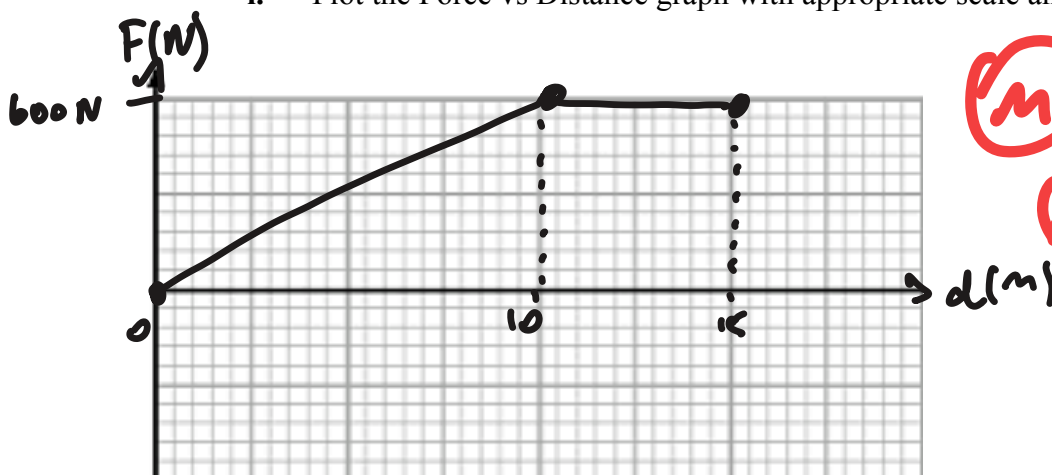
In practice, they conserve, in real life there is no closed body system possible.

Question 6 (6 Marks)

The house begins to invert. Gravity is no longer a constant 9.81 ms^{-2} , it is being sucked into a singularity into Peppa.

- a. A Force vs Distance graph for Peppa's pull on you shows the force F increases linearly from 0 N at $d = 0$ to 600 N at $d = 10 \text{ m}$, then remains constant at 600 N until $d = 15 \text{ m}$.
- i. Plot the Force vs Distance graph with appropriate scale and axis.

2 Marks



(M1) Scale, axis (M2) graph

- ii. Hence, calculate the total work done by Peppa on you over the 15m

1 Mark

$$W = Fd$$

$$W_1 = \frac{10 \cdot 600}{2} = 3000 \text{ J}$$

$$W_2 = 3000 \text{ J}$$

$$\therefore W_T = 6000 \text{ J}$$

- b. You are trying to climb a rope out of the singularity. You climb 4.0m in 2.0 seconds.

If your mass is 70kg, calculate the average power you must generate.

2 Marks

$$P = \frac{W}{t} = \frac{E_g}{t} = \frac{mgh}{t} = \frac{70 \cdot 9.81 \cdot 4.0}{2.0} = 1373 \text{ W or } 1.37 \text{ kW}$$

- c. If Peppa reaches the 'Event Horizon' and her velocity is 30 ms^{-1} with a momentum of 6000 kgms^{-1} , but she then 'glitches' and her kinetic energy suddenly doubles while her mass stays the same, calculate her new velocity.

3 Marks

$$E_{K_i} = \frac{1}{2}mv^2$$

$$= \frac{1}{2}(200)(30^2) = 90000 \text{ J}$$

$$E_{K_f} = 180000 \text{ J}$$

$$180000 = \frac{1}{2}mv^2$$

$$\sqrt{180000} = v$$

$$\therefore v = 42.4 \text{ ms}^{-1}$$

THE HILL HOUSE COLLAPSES INTO A SINGLE, PINK POINT OF LIGHT. YOU
ARE IN THE FOREST.

YOU ARE SAFE... FOR NOW...

END OF TEST